FIELD PROJECT REPORT

(Semester III – B.Sc. Data Science)



University of Mumbai

PROJECT IDENTIFICATION

ANALYSIS OF
SEASONAL DISEASE PATTERNS
USING GOVERNMENT HEALTH DATA

A study of NCDC / IDSP outbreak-surveillance records in Maharashtra

B.Sc. DATA SCIENCE | SEMESTER III | FIELD PROJECT

SUBMITTED BY	**Khan Umar Abdulaziz**
ROLL NUMBER	**41**
CLASS	**S.Y. B.Sc. Data Science**
SEMESTER	**III**
INSTITUTION	**R.P. Institute of Hospitality & Management**
UNIVERSITY	**University of Mumbai**
ACADEMIC YEAR	**2026-2027**
PROJECT GUIDE	**Prof. Angelin**

R. P. Institute
of Hospitality & Management
Vasai (W), Dist. Palghar, Maharashtra
Affiliated to University of Mumbai

University of Mumbai

CERTIFICATE

This is to certify that **Mr. Khan Umar Abdulaziz,**
Roll No. 41, a student of S.Y. B.Sc. (Data Science), Semester III,
R. P. Institute of Hospitality & Management, Vasai,
affiliated to the University of Mumbai, has successfully
completed the field project entitled

"Analysis of Seasonal Disease Patterns Using Government Health Data"

under the guidance of **Prof. Angelin,** during the academic year
2026 – 2027. This project is submitted in partial fulfillment of the
requirements for the Semester III examination of the
B.Sc. (Data Science) programme.

Prof. Angelin
Project Guide

College Stamp
Here

Prof. Khushali Sharma
Head of the Department

R. P. Institute
of Hospitality & Management
Vasai (W), Dist. Palghar, Maharashtra
Affiliated to University of Mumbai

University of Mumbai

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me during the completion of this field project.

I am deeply thankful to **Prof. Angelin**, my project guide, for her valuable suggestions, continuous encouragement and guidance throughout this project. Her support and insights have been instrumental in shaping this work.

I also extend my sincere thanks to **Prof. Khushali Sharma**, Head of the Department, for her constant motivation, valuable advice and support.

I am grateful to R. P. Institute of Hospitality & Management, Vasai, for providing the necessary resources and a conducive environment to undertake this project. I also acknowledge the University of Mumbai for the curriculum and the opportunity to work on this field project.

Lastly, I would like to thank my classmates, friends and family for their direct and indirect support during the course of this project.

DECLARATION

I, Khan Umar Abdulaziz, a student of S.Y. B.Sc. Data Science, Semester III, Roll No. 41, at R.P. Institute of Hospitality and Management, declare that I have prepared this field project report titled “Analysis of Seasonal Disease Patterns Using Government Health Data” for academic submission during the academic year 2026–2027.

I conducted the quantitative component using the NCDC/IDSP weekly outbreak-surveillance records and the analytical files maintained in my project repository. I have described the observation unit, data processing, statistical methods, findings and limitations so that the reader can distinguish government-reported outbreak records from individual patient counts and population incidence.

For the field component, I document the site context, key-informant questions and answers recorded in my field notes, and the observational details included in the project material. I have avoided identifying the informant by name and have treated photographs of public-health settings with care for privacy.

I have acknowledged the government data sources, repository outputs and background literature used in this report. The report is intended for academic evaluation and does not constitute clinical advice or a claim of causal proof about weather and disease transmission.

Student: Khan Umar Abdulaziz | Roll No.: 41
Place: Vasai, Maharashtra | Academic Year: 2026–2027
Student Signature: _____

ABSTRACT

Seasonal disease surveillance can help public-health teams prepare staff, diagnostics, vector-control activity and water-safety measures before reported outbreaks intensify. In this project, I analysed weekly outbreak-surveillance reports attributed to the National Centre for Disease Control (NCDC) and the Integrated Disease Surveillance Programme (IDSP) for Maharashtra. The main baseline covers the four completed calendar years 2022–2025 and contains 539 outbreak records across five selected disease families: dengue, acute diarrheal disease (ADD), malaria, food poisoning and chikungunya. The family-level tables sum to 20,955 reported cases and 266 reported deaths. I treated 2026 epidemiological Weeks 1–32 as a separate, partial comparison rather than a complete year.

I reviewed the repository’s data extraction, validation and disease-label normalization workflow, then compared monthly event distributions, seasonal concentration, year-to-year

stability, district distribution and sensitivity to influential records. Dengue reporting was concentrated from June to October, with October the largest monthly event count. ADD showed a June–August elevation, while malaria had a concentrated May–July window and was geographically clustered in eastern Vidarbha. Food-poisoning case totals were strongly affected by three high-volume February records. Chikungunya records were relatively sparse and did not support a stable statewide seasonal pattern.

I also documented a field visit to a public-health facility in the Nalasopara/Vasai–Virar area, including a key-informant interview and site observations, and related those qualitative notes to the limits of weekly government reporting. The study describes notified outbreaks rather than all illness in the population; it does not estimate incidence or prove that weather alone caused the observed patterns.

Keywords: seasonal surveillance, Maharashtra, NCDC, IDSP, dengue, malaria, data science, outbreak reporting, field observation.

INDEX / TABLE OF CONTENTS

The main report is organized around the project requirements: introduction, literature review, problem statement, objectives, methodology, data collection, results and analysis, field investigation, conclusion, references and appendices.

No.	Section / Chapter	Page
1	**1. Introduction**	4
2	**2. Literature Review**	5
3	**3. Problem Statement and Problems Faced**	6
4	**4. Research Objectives and Scope**	7
5	**5. Methodology and Analytical Framework**	8
6	**6. Data Collection and Processing**	10
7	7. Analysis and Results	11
8	7.1 Baseline Composition	11
9	7.2 Dengue	12
10	7.3 Acute Diarrheal Disease	13
11	7.4 Malaria	13
12	7.5 Food Poisoning	13
13	7.6 Chikungunya	14
14	7.7 Comparative Seasonal Profile	15
15	7.8 Annual Variation	16

No.	Section / Chapter	Page
16	7.9 Statistical Interpretation and Evidence Strength	16
17	7.10 Partial 2026 Comparison	16
18	**8. Fieldwork Investigation and Key Informant Q&A**	17
19	**9. Conclusion and Recommendations**	20
20	**10. References and Bibliography**	22
21	**11. Project Planner**	23
22	**12. Guide Interaction Diary**	23
23	**13. Appendices**	24

LIST OF FIGURES AND TABLES

• Figure 1. Research and data-processing workflow
• Figure 2. End-to-end data-processing pipeline
• Figure 3. Baseline outbreak records by disease family
• Figure 4. Share of the 539 baseline outbreak records by disease family
• Figure 5. February food-poisoning case volume before and after the three-record sensitivity check
• Figure 6. Monthly event counts for the five disease families, 2022–2025
• Figure 7. Three-month concentration for events compared with concentration for cases
• Figure 8. Annual reported outbreak records by disease family
• Figure 9. Partial 2026 family-level records and case totals through Week 32
• Figure 10. Field-site photographic documentation

List of Tables

• Table 1. Research scope and observation unit
• Table 2. Baseline disease-family totals
• Table 3. Seasonal timing and peak windows
• Table 4. Statistical concentration and stability
• Table 5. Outlier sensitivity results
• Table 6. Partial 2026 family-level totals
• Table 7. Field observation summary
• Table 8. Recommendation and seasonal lead time

CHAPTER 1

1. INTRODUCTION

1.1 Background and Public-Health Rationale

Seasonal disease surveillance is important because the timing of reported outbreaks affects when public-health teams prepare diagnostic supplies, medicines, vector-control activity, risk communication and safe-water measures. Maharashtra includes dense urban corridors, peri-urban settlements, coastal areas, agricultural districts and forested regions. A statewide total can therefore conceal meaningful differences in timing and geographic concentration.

The south-west monsoon usually changes local environmental conditions through rainfall, humidity, water accumulation, drainage pressure and household storage practices. These conditions can coincide with changes in mosquito-borne and gastrointestinal illness. However, this study does not assume that any single weather event caused a reported outbreak. It asks a narrower, answerable question: how are officially notified outbreak records distributed over months and years, and how reliable are the resulting patterns?

1.2 What the Surveillance Records Represent

The central observation unit in my analysis is one government-reported outbreak record. A record is not necessarily one patient, one laboratory-confirmed infection or a complete count of all symptomatic people in a locality. It reflects an event that entered the formal notification system. Mild illness managed at home, unreported private-clinic visits and cases that never trigger a formal outbreak investigation may not be represented. Consequently, the totals and percentages in this report describe reporting patterns, not population incidence, prevalence or individual risk.

1.3 Aim and Research Questions

My aim was to identify and compare seasonal patterns in five selected disease families using the government-reported outbreak records for Maharashtra. I asked: (1) which months contain the highest number of reported outbreak events for each family; (2) how stable these month rankings are across 2022–2025; (3) whether unusually large outbreak records distort case-volume peaks; (4) how disease records are distributed geographically; and (5) how field observations at a local public-health facility help explain the difference between daily clinical activity and weekly outbreak reporting.

1.4 Scope and Significance

The quantitative baseline includes four complete calendar years, 2022–2025. I kept the available 2026 records through Week 32 separate because later monsoon and post-monsoon weeks are missing from that comparison. The five primary families are dengue, acute diarrheal disease (ADD), malaria, food poisoning and chikungunya. The analysis is intended as a reproducible student data-science project and a practical description of outbreak-notification timing; it is not a clinical risk ranking or a forecasting model.



Figure 1. Research and data-processing workflow documented in the project.

CHAPTER 2

2. LITERATURE REVIEW

2.1 Seasonal Disease Patterns and Environmental Context

Public-health literature describes how disease patterns can vary with temperature, rainfall, humidity, water infrastructure, vector ecology and human behaviour. Dengue and chikungunya are associated with Aedes mosquitoes, while malaria transmission involves Anopheles mosquitoes. Enteric diseases and foodborne outbreaks follow different pathways, including contaminated water, unsafe storage and discrete shared-food exposures. These mechanisms provide context for studying seasonality, but they cannot be proved from a monthly table alone.

Bhatt et al. (2013) mapped the global distribution and burden of dengue and showed the value of combining geographic and epidemiological evidence. Gubler (2002) discussed dengue as a public-health problem shaped by urbanization and social conditions. Curriero et al. (2001) examined associations between extreme precipitation and waterborne-disease outbreaks. I use these studies to frame plausible context, not to claim that their estimates apply directly to the Maharashtra notification dataset.

2.2 Surveillance and Administrative Reporting

The Integrated Disease Surveillance Programme (IDSP), under the National Centre for Disease Control (NCDC) and the Ministry of Health and Family Welfare, is a key source of routine outbreak information in India. Weekly reports allow events to be examined by reporting period, district, disease label and reported case count. They are operational surveillance documents, not a person-level census. The distinction matters because the number of outbreak notifications can remain stable while the number of cases attached to an event changes substantially.

2.3 Statistical Approaches Relevant to This Project

Seasonal indices help identify months with above-average reporting. Concentration ratios summarize the proportion of events or cases contained within selected months. The coefficient of variation and Gini coefficient describe dispersion and inequality across monthly totals. Kendall's coefficient of concordance (W) assesses agreement among annual month-rank profiles, while a Kruskal–Wallis test assesses whether values differ across month groups under its assumptions. Each measure answers a different question; no single statistic is sufficient to certify a biological seasonal cycle.

2.4 Research Gap

The practical gap addressed by this project is the effort required to transform scattered weekly PDFs into an auditable, multi-year dataset that distinguishes repeated temporal structure from label variation, missing reports, district concentration and a few exceptional high-case events. I also sought to connect the quantitative findings with a local field visit, while maintaining a clear distinction between state-level surveillance patterns and local observations.

CHAPTER 3

3. PROBLEM STATEMENT AND PROBLEMS FACED

3.1 Problem Statement

Weekly disease-outbreak bulletins are published as PDFs rather than as one consistently structured analysis table. Before analysis, the source material must be extracted, checked against its original report, and normalized. If this work is skipped, spelling variants may split the same disease into separate categories, incomplete time periods may be compared unfairly, and isolated high-case events may dominate the apparent seasonal profile. This project

responds to that problem by documenting a controlled processing workflow and comparing event frequency with reported case volume.

3.2 Practical Problems Encountered

Problem	How I handled it
PDF extraction and table layout	Weekly bulletins required document-level extraction and manual review where line breaks or table formatting could alter a field.
Disease-label variation	Different spellings and related labels needed harmonization without discarding their original values.
Missing source weeks	The 2023 archive lacks published Weeks 15, 51 and 52. I did not create artificial zero-event records for absent reports.
Partial-year comparison	The 2026 dataset ends at Week 32 and omits later weeks when several disease families historically remain active.
Event count versus case volume	One large food-poisoning cluster can affect case totals much more than event counts; both measures were retained.
Uneven geographic reporting	Differences between districts may reflect burden, access, definitions and surveillance intensity as well as underlying disease dynamics.

3.3 How I Addressed These Problems

I maintained separate raw and normalized disease labels, preserved record provenance, compared outputs across analytical phases, kept the 2022–2025 baseline separate from partial 2026, and performed sensitivity checks on influential records. I also interpreted statistical significance cautiously because only four completed annual profiles are available for inter-annual comparisons.

CHAPTER 4

4. RESEARCH OBJECTIVES AND INQUIRY SCOPE

4.1 Objectives

No.	Objective	Project outcome
1	Assemble and validate Maharashtra outbreak records from weekly NCDC/IDSP reports.	Completed in repository analysis
2	Normalize disease labels and select five primary families for comparative analysis.	Completed
3	Describe monthly and annual outbreak-reporting distributions and identify peak windows.	Completed
4	Compare concentration and year-to-year stability across the 2022–2025 baseline.	Completed
5	Test the sensitivity of case-volume peaks to influential outbreak records.	Completed

No.	Objective	Project outcome
6	Compare partial 2026 Week 1–32 reporting with prior equivalent periods.	Completed with temporal boundary
7	Document local field context through a key-informant interview and site observations.	Field notes and site photographs documented
8	Translate the analysis into practical, evidence-led recommendations.	Completed

4.2 Study Area and Unit of Analysis

The quantitative study covers Maharashtra, including records attributed to its 36 districts. The field visit was focused on a public-health facility in the Nalasopara/Vasai–Virar area of Palghar district. These geographic scopes are not interchangeable: the statewide dataset describes reported events across the state, while the field component provides a local view of facility operations and surrounding infrastructure.

Dimension	Specification
Quantitative geography	Maharashtra state, with district-level comparison
Primary period	2022–2025 completed calendar years
Secondary comparison	2026 epidemiological Weeks 1–32
Observation unit	One government-reported outbreak record
Primary families	Dengue, ADD, malaria, food poisoning, chikungunya
Field setting	Public-health facility and surrounding setting in Nalasopara / Vasai–Virar, Palghar

4.3 Delimitations

I did not estimate disease incidence per 100,000 population, conduct clinical diagnosis, or model weather variables as causal predictors. I treated field observations as local context and did not use them to calculate statewide rates. The 2026 partial-year comparison is not presented as a full-year forecast.

CHAPTER 5

5. METHODOLOGY AND ANALYTICAL FRAMEWORK

5.1 Research Design

I used a quantitative, descriptive and analytical design for the government data, supported by qualitative context from a key-informant interaction and a structured site walk-through. The strands serve different purposes. The data analysis identifies statewide patterns in notified

events; the field component helps describe how a local facility sees illness and processes information. Local observations do not alter the statewide event counts.

5.2 Data Processing Stages

Stage	Work undertaken	Output
1. Source review	Organized weekly NCDC/IDSP PDFs by year and epidemiological week.	Archived source reports
2. Extraction	Extracted Maharashtra outbreak rows and associated fields.	809 extracted records across all disease families
3. Provenance and validation	Checked extracted rows against source reports and retained validation metadata.	Row-level audit trail
4. Harmonization	Mapped inconsistent disease names to canonical labels and analytical families.	30 analytical disease families
5. Cohort isolation	Selected five families with sufficient baseline observations for seasonality analysis.	539 records for 2022–2025
6. Aggregation	Aggregated records by month, epidemiological week, year and district.	Seasonal and spatial summary tables
7. Analytical tests	Calculated concentration metrics, seasonal indices, stability tests and sensitivity analyses.	Phase 5B–5D outputs
8. Interpretation	Compared findings and documented limitations before writing recommendations.	Evidence-graded conclusions



Figure 2. End-to-end data pipeline: from weekly reports to evidence-led interpretation.

5.3 Statistical Measures

Measure	Purpose in my analysis
Monthly event count	Counts report records assigned to each calendar month.
Seasonal index	Compares each month's event frequency with the average monthly level; values above 1 indicate above-average reporting.
Three-month concentration ratio (CR3)	Measures the share of events captured within the selected three-month window.
Coefficient of variation (CV)	Measures relative dispersion of monthly counts around their mean.
Gini coefficient	Measures inequality in the distribution of monthly event counts.
Kruskal–Wallis test	Tests for differences in distribution across month groups; the result is interpreted alongside assumptions and sample size.

Measure	Purpose in my analysis
Kendall's W	Measures agreement among the four annual month-rank profiles; the p-value is treated as exploratory.
Outlier sensitivity	Recalculates monthly case patterns after excluding a small number of influential event records, solely as a sensitivity check.

5.4 Field Instrument and Ethics

For the field component I used a semi-structured key-informant interview format and a structured observation approach. I kept the informant anonymous, did not reproduce identifiable patient records, and used photographs for public-facility context rather than individual identification. The field questions focused on seasonal timing, diagnostic workflow, outbreak response, foodborne versus diarrheal presentations, reporting and opportunities to improve information flow.

CHAPTER 6

6. DATA COLLECTION AND PROCESSING PIPELINE

6.1 Primary Government Source

The core source is the NCDC/IDSP weekly outbreak surveillance archive published under the Ministry of Health and Family Welfare, Government of India. The repository's source audit records weekly reports from Week 25 of 2022 through Week 32 of 2026, with the analytical dataset covering 809 extracted records across multiple disease families. Within this archive, the five selected families form the primary quantitative cohort.

6.2 Dataset Profile

Metric	Value used in this report
Extracted records across all disease families	809
Primary disease-family records, 2022–2025	539
Reported cases in five baseline family tables	20,955
Reported deaths in five baseline family tables	266
Primary disease families	5
Partial 2026 records in the five families, Weeks 1–32	45
2026 reported cases summed across the five family rows	2,126
2023 weeks missing from the published archive	15, 51 and 52

6.3 Disease Taxonomy and Selection

The extraction contained many variations in disease names. I retained source terminology for traceability and standardized the analysis labels so that variants such as “Dengue” and “Dengu”, or “ADD” and expanded diarrheal-disease spellings, could be grouped consistently. The main five-family cohort was selected using the repository’s baseline event thresholds and relevance to seasonal comparison.

6.4 Data Quality Decisions

I did not fill missing report weeks with artificial zeroes. I retained high-volume records in the baseline and then examined their influence through sensitivity analysis rather than deleting them from the source data. I also separated partial 2026 from complete years to avoid interpreting the absence of late-season weeks as a decline in annual disease activity.

The 2022–2025 family totals used consistently throughout the report are 204 dengue records, 153 ADD records, 70 malaria records, 69 food-poisoning records and 43 chikungunya records. These values sum to 539 records. Their case totals sum to 20,955 and the family death totals sum to 266.

CHAPTER 7

7. ANALYSIS AND RESULTS: FIVE DISEASE SEASONAL SIGNATURES

7.1 Baseline Composition

The baseline contains 539 outbreak records. Dengue contributes the largest number of records (204; 37.8%), followed by ADD (153; 28.4%). Malaria and food poisoning contribute 70 (13.0%) and 69 (12.8%) records respectively, while chikungunya contributes 43 (8.0%). Record share describes the distribution of notified events, not relative population risk.

Disease family	Records	Reported cases	Reported deaths	Share of records
Dengue	204	3,144	93	37.8%
Acute diarrheal disease	153	9,050	93	28.4%
Malaria	70	2,133	51	13.0%
Food poisoning	69	5,927	29	12.8%
Chikungunya	43	701	0	8.0%
TOTAL	539	20,955	266	100.0%

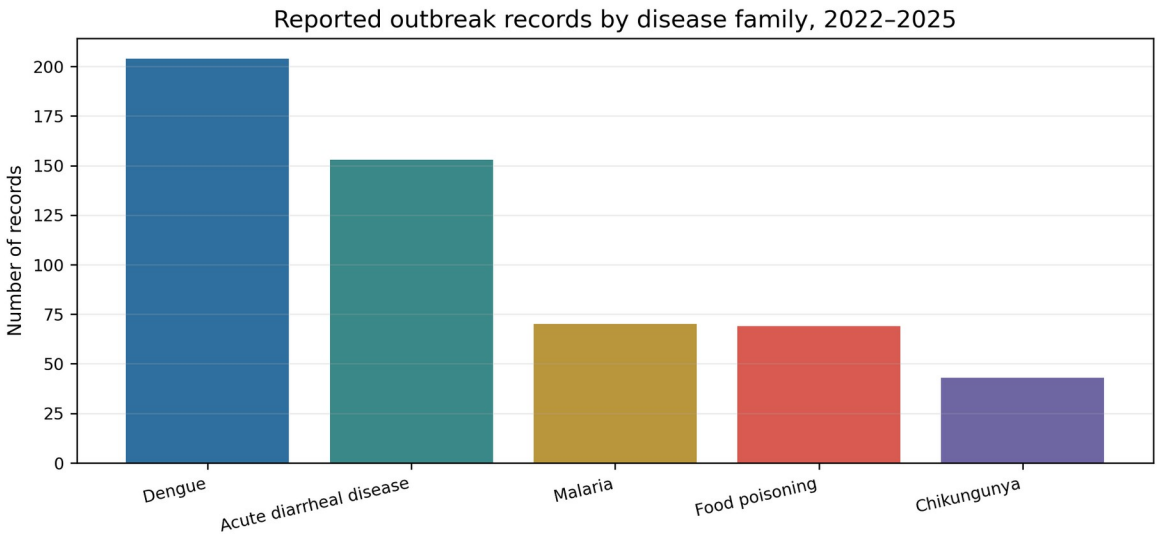


Figure 3. Baseline outbreak records by primary disease family.

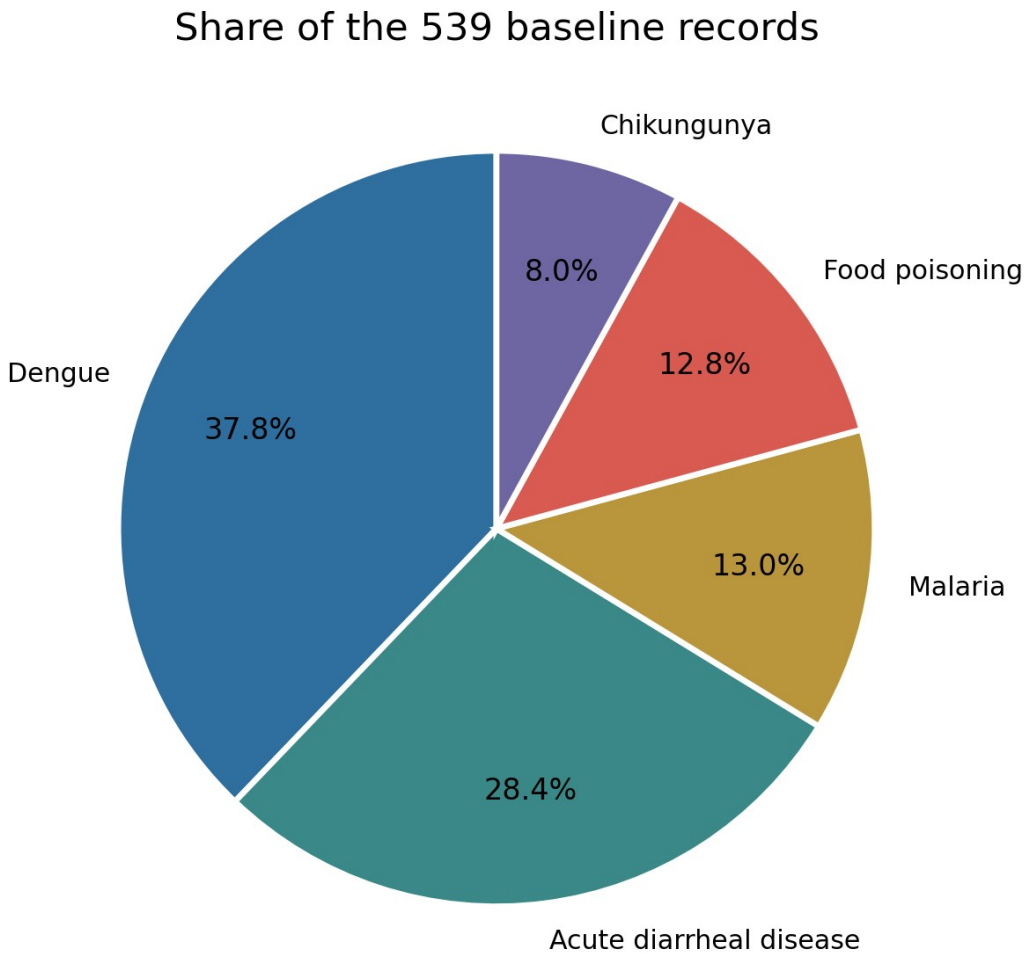


Figure 4. Share of the 539 baseline outbreak records by family.

7.2 Dengue: Strong Late-Year Concentration

Dengue is the largest baseline family, with 204 records, 3,144 reported cases and 93 reported deaths. The June–October window contains 169 of the 204 records (82.8%). October is the

peak notification month, with 40 records (19.6%); September follows with 36 (17.6%). The high concentration extends across several adjacent months, rather than depending on one isolated outbreak record.

The repository reports strong monthly heterogeneity for dengue (Kruskal–Wallis p = 0.000219). Kendall's W is 0.405 with p = 0.086, indicating moderate but not statistically significant annual rank concordance under the reported approximation. Dengue reporting also appears across 31 districts, suggesting that the pattern is not limited to one reporting district. The most defensible finding is descriptive: the baseline reports a strong June–October concentration, with the largest event count in October.

7.3 Acute Diarrheal Disease: Mid-Year Elevation

ADD is the largest family by reported case volume: 153 records, 9,050 cases and 93 reported deaths. June through August contain 66 records (43.1%). June's seasonal index is 1.49, July's is 1.73 and August's is 1.25, with a secondary October elevation (25 records, 16.3%). The family is widely distributed across 33 districts in the repository analysis.

Kendall's W is 0.530 (p = 0.016), the strongest inter-annual rank concordance among the five families. The repository's Kruskal–Wallis result is not significant (p = 0.1321), so the evidence should not be summarized as one test proving all forms of seasonality. The appropriate interpretation is a recurring descriptive June–August elevation with exploratory support for similarity in monthly ranking across years.

7.4 Malaria: May–July Concentration with a Regional Footprint

Malaria contributes 70 records, 2,133 reported cases and 51 reported deaths. The May–July window contains 36 records (51.4%), and June (15 records; 21.4%) and July (14; 20.0%) are the largest monthly counts. The Kruskal–Wallis p-value is 0.005459, while Kendall's W is 0.319 (p = 0.229), indicating that monthly variation is present but year-to-year rank agreement is more modest.

Geographically, Gadchiroli contributed 27 events and Chandrapur 16. Together they account for 43 of 70 malaria records (61.4%). This concentration matters: the aggregate Maharashtra curve is heavily influenced by eastern Vidarbha and should not be assumed to represent every district uniformly.

7.5 Food Poisoning: Bimodal Events and Outlier-Sensitive Cases

Food poisoning contributes 69 records, 5,927 reported cases and 29 reported deaths. Events are distributed in two broader periods: January–February contributes 26.1% of records and

April–May contributes 29.0%. Unlike a sustained seasonal wave, food-poisoning notifications can represent a common exposure affecting many people at once.

The difference between event count and case volume is especially important. Three February records from Kolhapur (651 cases), Parbhani (629) and Solapur (335) contribute 1,615 cases. Removing these records for sensitivity analysis reduces February's case total from 2,162 to 547 and its share from 36.5% to 12.7%; April becomes the highest remaining case month with 899 cases. I retain the records in the main dataset and use the exclusion only to show how much a few high-volume outbreaks affect the monthly case profile.

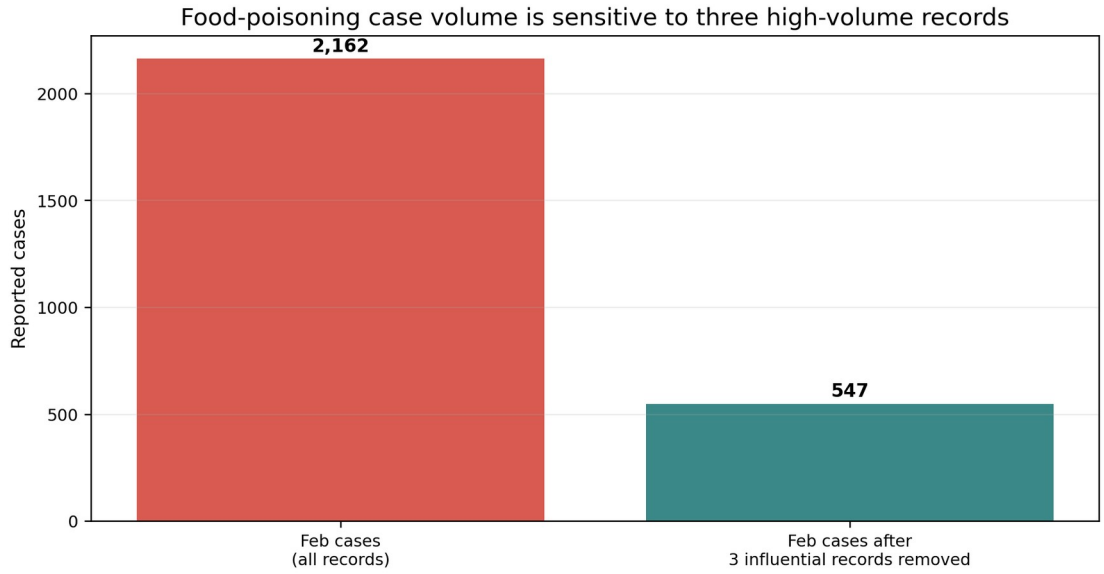


Figure 5. February food-poisoning case volume before and after the three-record sensitivity check.

7.6 Chikungunya: Sparse and Diffuse Evidence

Chikungunya has 43 baseline records, 701 reported cases and no reported deaths in the selected family table. The leading months include May and June (six events each), but no stable contiguous peak window is established. The three-month event concentration ratio is 39.5%, and Kendall's W is 0.150 ($p = 0.832$). Pune contributed 11 of 43 records (25.6%). These data are too sparse and unstable to support a strong statewide seasonal claim.

7.7 Comparative Seasonal Profile

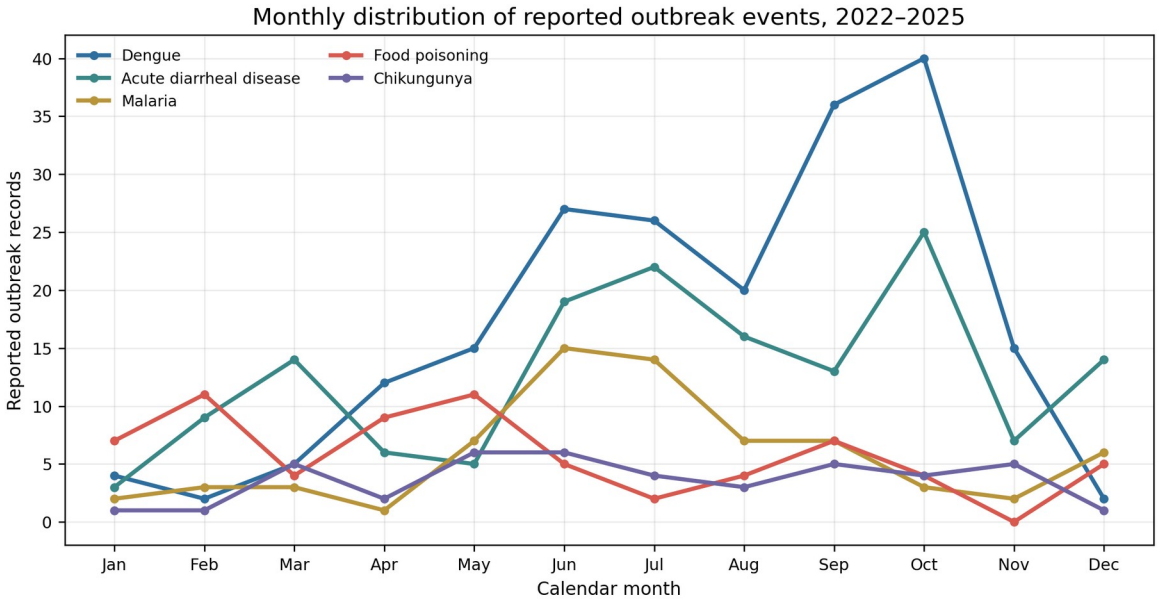


Figure 6. Monthly event counts for the five disease families, 2022–2025.

Family	Most informative timing pattern	CR3 (events)	Kendall's W (p)	Interpretation
Dengue	June–October; October peak	50.5%	0.405 (0.086)	Strong descriptive concentration
ADD	June–August elevation; October secondary rise	43.1%	0.530 (0.016)	Moderate descriptive pattern; W is exploratory
Malaria	May–July; June peak	51.4%	0.319 (0.229)	High temporal concentration, geographically focal
Food poisoning	January–February and April–May	44.9%	0.386 (0.108)	Variable, strongly case-volume sensitive
Chikungunya	Diffuse; no stable contiguous window	39.5%	0.150 (0.832)	Insufficient evidence for stable statewide timing

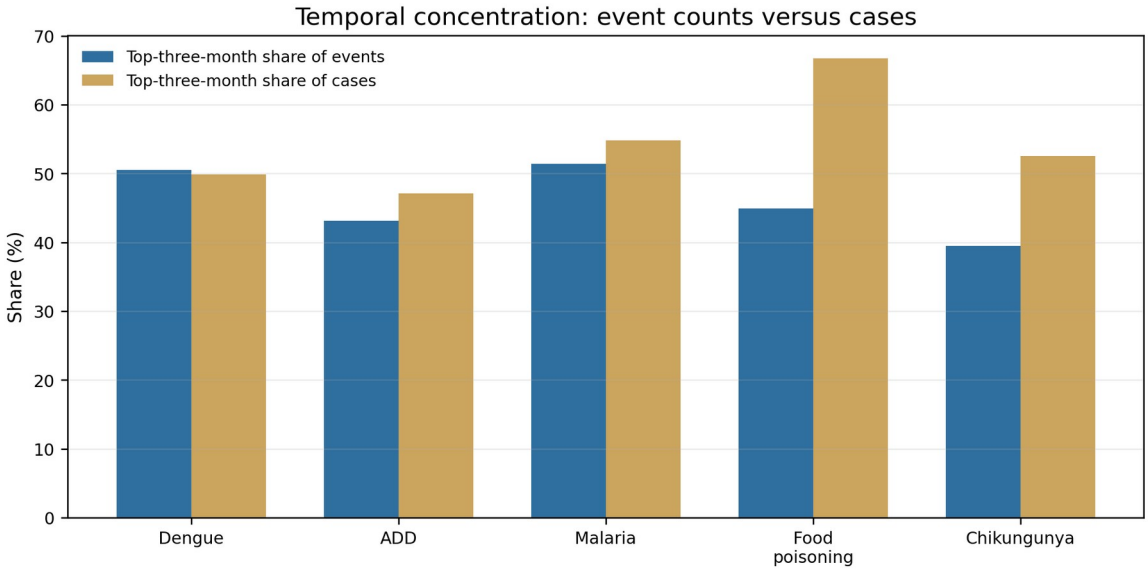


Figure 7. Three-month concentration for events compared with concentration for cases.

7.8 Annual Variation

The annual profiles show that the same number of years can contain very different event volumes. Dengue reported 20 events in 2022, 104 in 2023, 63 in 2024 and 17 in 2025. ADD rose to 71 records in 2024; food poisoning reached 29 in 2025; malaria stayed between 11 and 22; chikungunya's 22 records in 2024 represented more than half of its four-year baseline. These shifts are a reason to separate annual totals from a statement of predictable biological seasonality.

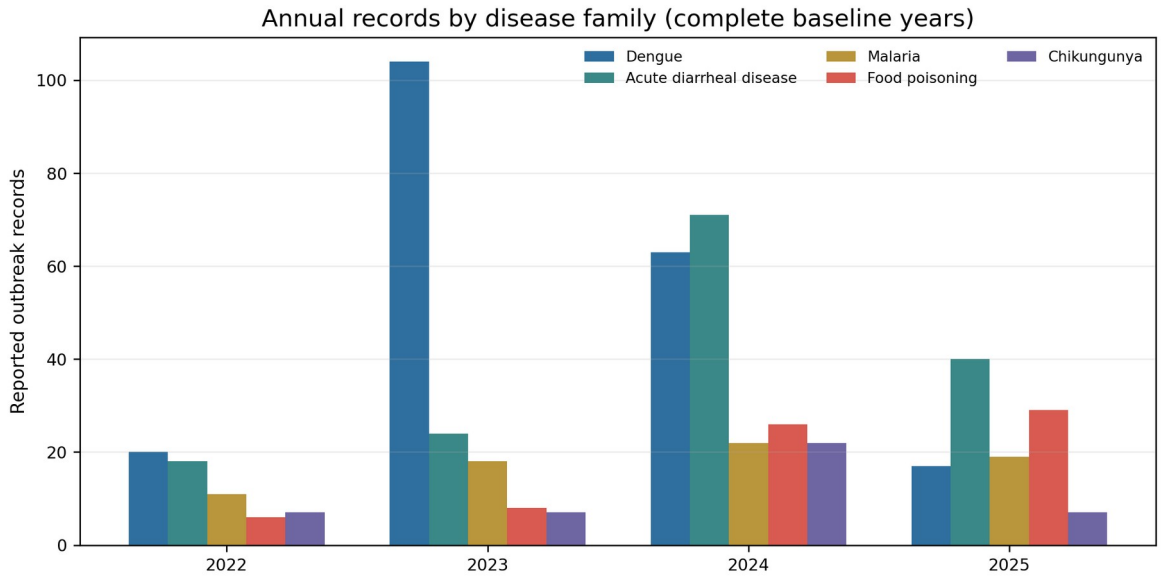


Figure 8. Annual reported outbreak records across the four complete baseline years.

7.9 Statistical Interpretation and Evidence Strength

The concentration statistics and monthly tests answer different questions. A high CR3 means events are clustered in a short window; it does not prove that the same peak repeats every year. A significant Kruskal–Wallis result indicates monthly heterogeneity under the test design; it does not prove climate causation. Kendall's W describes agreement among annual monthly ranks, but the project has only four annual profiles and uses an approximate chi-square p-value. I therefore interpret the tests together with event counts, district coverage, year-to-year changes and sensitivity results.

7.10 Partial 2026 Comparison through Week 32

I analysed the first 32 epidemiological weeks of 2026 separately from the completed-year baseline. The five family rows contain 45 records: six dengue, 18 ADD, one malaria, 19 food-poisoning and one chikungunya record. The corresponding family-level case totals sum to 2,126. This window can describe activity up to Week 32 only. It does not include the full

late-monsoon/post-monsoon period, so the lower dengue and malaria event counts cannot be read as full-year reductions.

Family	Records (2026 W1–32)	Reported cases	What the partial period shows
Dengue	6	106	Late-season period is incomplete; no full-year conclusion
ADD	18	864	Several records across the partial period; still not a full-year comparison
Malaria	1	65	One reported event in the partial table; investigate locally, do not infer statewide absence
Food poisoning	19	1,082	Highest record count among five families in this partial window
Chikungunya	1	9	Too few records to infer a seasonal pattern
TOTAL	45	2,126	Partial-year comparison only

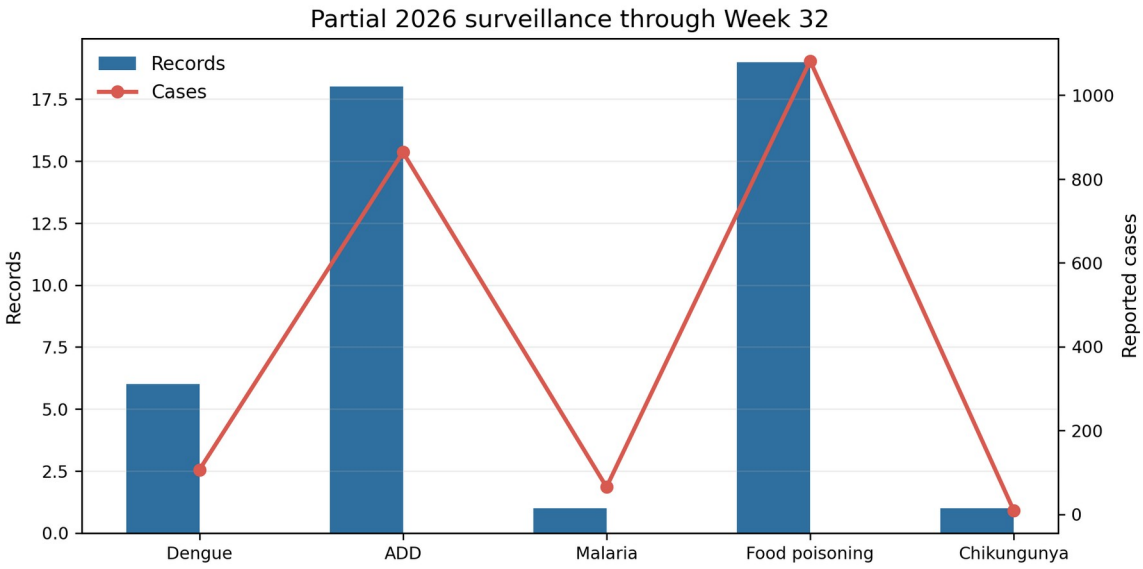


Figure 9. Partial 2026 family-level records and case totals through Week 32.

CHAPTER 8

8. FIELDWORK INVESTIGATION AND KEY-INFORMANT QUESTIONNAIRE

8.1 Field Visit and Facility Context

I visited the public-health facility in the Nalasopara West / Vasai–Virar area of Palghar district to understand how seasonal illness is encountered at facility level and how operational observations relate to the state surveillance data. The facility environment included a patient waiting corridor, a reception/triage area and public-facing facility infrastructure. I recorded

the visit as contextual qualitative work, not as a representative survey of the entire population of Maharashtra.

8.2 Structured Key-Informant Questionnaire: Questions and Answers

I used a semi-structured question set with the Medical Officer-in-Charge. The questions were arranged around seasonal disease timing, diagnostic processes, reporting pathways, outbreak response, foodborne illness and potential improvements to data flow. The answers below are paraphrased from the interview field notes included in my project draft; the informant is not named.

Q1. Seasonal shifts: What seasonal shifts in patient volume and disease presentation do you observe throughout the year?

Recorded answer: The recorded response described two broad periods: gastrointestinal illness and diarrhoea increasing around the June–July monsoon onset, and fever presentations—including suspected dengue and viral fever—becoming more prominent from September to November.

Q2. Vector ecology: How does local vector-borne transmission differ from rural or forested districts?

Recorded answer: The response distinguished dense peri-urban Nalasopara from forested districts elsewhere in Palghar. It associated local risk with construction-related standing water and household/peri-household containers such as storage drums and overhead cisterns.

Q3. Diagnostic workflow: How are suspected outbreak diseases diagnosed and recorded at the facility?

Recorded answer: The response described clinical assessment and rapid diagnostic tests, including malaria antigen cards and dengue NS1/IgM tests. Where indicated, samples are sent to a sub-district or district reference laboratory for further confirmation.

Q4. Reporting hierarchy: How does information move from outpatient registration to IDSP reporting?

Recorded answer: The response described daily entries in physical outpatient registers, consolidation of fever and diarrhoea information, classification using presumptive or laboratory-confirmed reporting streams, and periodic submission to the Palghar District Surveillance Unit.

Q5. Containment triggers: What happens when several cases cluster in one neighbourhood?

Recorded answer: The response described notifying municipal sanitation teams, checking water-storage practices, distributing chlorine tablets, conducting mosquito-control activity and arranging oral rehydration support when a local cluster is identified.

Q6. Logistical hurdles: What difficulties arise during seasonal peaks?

Recorded answer: The response highlighted pressure on rapid-test supplies during sudden fever surges, delays in confirmatory laboratory results and delayed presentation to public facilities when people first self-medicate or consult private pharmacies.

Q7. Clinical presentation: How does food poisoning differ from general acute diarrhoeal disease?

Recorded answer: The response distinguished sudden groups of people becoming ill after a shared meal or gathering from diarrhoeal cases appearing across households over a longer period.

Q8. Summer heat: Does hot weather affect foodborne or gastrointestinal presentations?

Recorded answer: The response identified hotter months as a period when unsafe food handling, unrefrigerated dairy and stored drinking water may create additional food-safety concerns. This is recorded as the informant's operational perspective, not as a causal estimate from my dataset.

Q9. Surveillance gaps: What is missed between daily clinical activity and official weekly reports?

Recorded answer: The response explained that official bulletins aggregate investigated outbreak clusters, while routine mild cases may remain in facility registers and not appear as separate outbreak notifications unless the cluster meets reporting criteria.

Q10. Technology: What change would improve facility-level surveillance?

Recorded answer: The response favoured direct digital entry at the triage point, automated alerts when reported counts rise, and better integration between facility entries and reference-laboratory results to reduce duplicate transcription.

8.3 Structured Site Observations

I recorded three main observations during the facility walk-through. They describe the setting and workflow visible during one visit; they are not a statistical sample of patients or a formal facility audit across multiple dates.

Time / zone	Observation recorded
10:40 AM — Outpatient corridor	High morning activity was visible in the waiting corridor, and patient triage appeared to rely on manual workflow. My field note estimated that a substantial portion of people waiting had fever or gastrointestinal complaints; this was a one-time visual estimate and not a measured clinical statistic.
11:15 AM — Point-of-care area	The field note described an accessible oral rehydration area and rapid diagnostic-test materials for malaria and dengue. The observation was limited to visible site context and did not constitute an independent stock audit.
12:00 PM — Facility perimeter and nearby housing	The field note described roadside drainage with intermittent stagnant water and domestic water storage in large plastic drums in the surrounding residential setting. The potential significance is as a context for environmental-health questions, not as proof that these features caused a specific reported outbreak.

8.4 What the Field Visit Added to the Data Analysis

The most useful contribution of the visit was contextual rather than statistical. The interview notes describe how daily clinical activity may be more detailed than the weekly outbreak bulletin, because the bulletin captures notified clusters rather than all outpatient visits. The

reporting pathway—from facility registers to district-level aggregation—also clarifies why a weekly publication date is not identical to the date an illness first appeared.

The field notes also help interpret the difference between ADD and food poisoning. A prolonged spread of gastrointestinal illness across households creates a different reporting pattern from a single shared exposure that produces a large cluster in a short period. This is consistent with the quantitative sensitivity analysis, in which a few large food-poisoning records change the monthly case-volume distribution substantially.

8.5 Link between Field Context and Statewide Surveillance

Field-context theme	Relevant quantitative finding	How I interpret the connection
Monsoon-associated gastrointestinal presentations	ADD records rise in June–August.	Local context is compatible with the timing, but does not prove a cause.
Late-monsoon and post-monsoon fever concerns	Dengue records concentrate June–October, with October peak.	The interview perspective is directionally consistent with the statewide notification profile.
Shared-meal clusters	Food-poisoning case totals are highly sensitive to three records.	Point-source episodes can increase case totals without a proportional increase in event frequency.
Daily register and weekly notification pathway	The study uses weekly outbreak PDFs.	Operational reporting thresholds and aggregation mean clinical workload and bulletin counts are different measures.
Standing water and household storage	Malaria is concentrated in eastern Vidarbha; dengue is widely distributed.	Local environmental context is relevant to prevention planning but is not an explanatory variable measured in this dataset.

CHAPTER 9

9. CONCLUSION AND SUGGESTIONS

9.1 Conclusion

Through this project, I transformed weekly NCDC/IDSP outbreak reports into a structured dataset and compared five disease families across the complete 2022–2025 baseline. The strongest descriptive pattern was dengue’s late-year concentration: 82.8% of its baseline events occurred from June to October, with October the largest month. ADD had a recurring June–August elevation, malaria events were concentrated in May–July and strongly represented by Gadchiroli and Chandrapur, food-poisoning case totals were dominated by several high-volume events, and chikungunya records were too sparse for a stable statewide seasonal conclusion.

The main methodological lesson is that the frequency of outbreak records and the number of reported cases are not interchangeable. A cluster with many cases can move the monthly case

curve while contributing only one event record. Therefore, I interpreted the event distribution, case distribution, statistical tests, district footprint, year-to-year variation and sensitivity analysis together. I also kept partial 2026 records separate from complete years so that unobserved late-season weeks would not be mistaken for declining annual activity.

The field visit added a local operational perspective on patient flow, rapid testing, municipal response and weekly reporting. It supports the importance of linking administrative data with the realities of facility workflow, but it does not establish statewide prevalence or causal effects. The project's overall contribution is a transparent, reproducible description of government-reported outbreak seasonality and a set of practical questions for future surveillance improvement.

9.2 Recommendations

Recommendation	Application
1. Prepare ahead of the early monsoon	Use the May–June period to review water-safety communication, ORS availability, outbreak triage and reporting readiness before the observed ADD elevation.
2. Sustain vector-control activity through late monsoon	Because dengue notifications are concentrated June–October, source reduction, household container management and public communication should extend beyond the first rains.
3. Use district-sensitive malaria planning	The high share of records from Gadchiroli and Chandrapur suggests that district-level planning is more informative than a uniform statewide interpretation.
4. Separate event alerts from case-volume alerts	A small number of mass-exposure food-poisoning events can affect case totals sharply; dashboards should display record counts and case counts side by side.
5. Improve data-entry and notification linkage	Explore digital support to reduce repeated transcription from facility records to district-level reporting, while preserving verification and audit trails.
6. Keep incomplete periods clearly bounded	Interim 2026 dashboards should show the last observed week prominently and avoid treating early-year counts as full-year outcomes.
7. Maintain consistent disease labels	Store raw source strings beside canonical family labels and review label shifts periodically to protect trend comparability.
8. Continue local field documentation	Future site visits can use anonymized questionnaires, observation checklists and consent-based notes to compare local perception with the surveillance calendar.

9.3 Limitations and Future Work

The records are outbreak notifications rather than individual case line lists and have no population denominator for incidence calculations. The 2023 source archive has three missing weekly bulletins, and the partial 2026 period excludes later weeks. Reporting intensity and case classification may vary by district and year. Kendall's W is estimated from only four annual profiles, and statistical significance does not establish a biological

mechanism or causal role for rainfall. The field component is localized to one facility setting and is not representative of all districts.

Future work could extend the baseline as newer weekly reports become available, re-run the analysis using a versioned machine-readable extract, compare notification timing with independent weather observations using a pre-specified lag structure, and expand fieldwork with documented consented responses across multiple settlement types. The next step for field research is to retain completed forms and anonymized notes in the evidence register so that qualitative summaries can be audited.

CHAPTER 10

10. REFERENCES AND BIBLIOGRAPHY

10.1 Government and Data Sources

1. National Centre for Disease Control (NCDC), Directorate General of Health Services, Ministry of Health and Family Welfare, Government of India. Weekly Outbreak Surveillance Reports / Integrated Disease Surveillance Programme (IDSP), Maharashtra records, 2022–2026. https://ncdc.mohfw.gov.in/

2. Integrated Disease Surveillance Programme (IDSP). Programme information and surveillance resources. Government of India. https://idsp.nic.in/

3. Ministry of Health and Family Welfare / IDSP. IDSP outbreak master archive, hosted by Dataful (national dataset and metadata). https://dataful.in/datasets/18514

4. Project repository. Field Project: Analysis of Seasonal Disease Patterns Using Government Health Data. https://github.com/RizzyCoder19/Field-Project

5. Project website. Seasonal disease pattern analysis in Maharashtra. https://fieldprojects.vercel.app/

6. Project repository artefacts used: Phase 5B seasonal-analysis report and monthly/yearly distribution tables; Phase 5C advanced-analysis and outlier-sensitivity tables; Phase 5D interpretation, validation and limitation documents; fieldwork methodology, questionnaire, interview guide, observation checklist and ethics protocol.

10.2 Peer-Reviewed Literature

7. Bhatt, S., Gething, P. W., Brady, O. J., Messina, J. P., Farlow, A. W., Moyes, C. L., et al. (2013). The global distribution and burden of dengue. Nature, 496, 504–507.

8. Curriero, F. C., Patz, J. A., Rose, J. B., & Lele, S. (2001). The association between extreme precipitation and waterborne disease outbreaks in the United States, 1948–1994. American Journal of Public Health, 91(8), 1194–1199.

9. Gubler, D. J. (2002). Epidemic dengue/dengue hemorrhagic fever as a public health, social and economic problem in the 21st century. Trends in Microbiology, 10(2), 100–103.

10. Kendall, M. G., & Babington Smith, B. (1939). The problem of m rankings. The Annals of Mathematical Statistics, 10(3), 275–287.

CHAPTER 11

11. PROJECT PLANNER / EXECUTION TIMELINE

I organized the project as a sequence of source review, data engineering, statistical analysis, interpretation, field contextualization and report preparation. The following planner records the main work stages and their outputs.

Stage	Work completed	Output
Weeks 1–2	Review source archive and repository; organize weekly PDFs and verify source fields.	Source register and extraction plan
Weeks 3–4	Extract Maharashtra rows, check provenance and normalize disease labels.	Validated master records and label map
Weeks 5–6	Build the baseline, monthly/annual profiles, concentration measures and stability tests.	Phase 5B and Phase 5C outputs
Week 7	Run outlier sensitivity checks, district comparisons and partial-2026 analysis.	Sensitivity tables and validation summary
Week 8	Complete field-context notes, recommendations, visualizations and report.	Report, figures, references and appendices

11.1 Project Deliverables

The final project package includes the research report, source and analytical files in the GitHub repository, the interactive project website, statistical charts, questionnaire and interview instruments, field-site photographs, and this written account of the methods and findings.

CHAPTER 12

12. GUIDE INTERACTION DIARY / SUPERVISION RECORD

I maintained the following topic-based record of the project's development and review focus. The entries describe the workstream and the action taken, rather than using a signature field as evidence of approval.

Date	Review focus	Action / learning recorded
12 Aug 2026	Topic selection, research questions and NCDC/IDSP surveillance scope.	Defined the Maharashtra focus and kept source provenance central to the project.
26 Aug 2026	Disease-label harmonization and preservation of original source strings.	Used additive normalization so that the source label remained available for audit.
10 Sep 2026	Seasonal concentration, CR3 and inter-annual concordance.	Compared descriptive concentration with the limits of statistical tests.
24 Sep 2026	Food-poisoning outlier sensitivity.	Retained influential records and added a sensitivity comparison rather than deleting data.
03 Oct 2026	Field context, final interpretation and report structure.	Linked the site visit to the surveillance analysis while keeping local observations distinct from statewide results.

CHAPTER 13

13. APPENDICES

Appendix A. Rapid Field Questionnaire

The following concise questionnaire is designed for a voluntary, anonymous conversation with a community member or local stakeholder. It can be answered through selected response options and does not request names, phone numbers, patient details or other identifying information.

1. In your experience, do acute illnesses in your locality increase during particular months?
2. Which period seems to have the highest illness activity: summer, early monsoon, late monsoon, post-monsoon or winter?
3. Which of the following have you observed or heard diagnosed locally: dengue, malaria, ADD, food poisoning, chikungunya?
4. In which months do mosquito-borne fevers seem most visible in your area?
5. When do diarrhoea or stomach infections seem to increase?
6. Have you noticed a localized group of people becoming ill after a common meal or gathering?
7. What environmental condition is most visible near the residence or workplace: stagnant drains, water-filled potholes, construction water, uncovered storage, or none seen?
8. What is the main drinking-water source: municipal tap, tank, tanker/standpost, packaged/filtered water or well/borewell?
9. How often do you notice mosquito-control or sanitation activity in the locality?
10. Have you seen public-health messages about preventing dengue, malaria or diarrhoeal disease?

11. If someone develops fever, where do they usually seek help first: public facility, private clinic, pharmacy or home care?
12. What usually encourages people to seek medical attention?
13. Are diagnostic tests easy to access locally?
14. Are public-health workers or community workers visible in the locality?
15. How quickly do residents hear about a local outbreak or prevention drive?
16. What is the most difficult part of getting care during an illness surge?
17. Have you noticed waterlogging or drainage problems after heavy rain?
18. What could the municipality improve before the monsoon?
19. Would simple early-warning messages by SMS, WhatsApp, posters or local announcements be useful?
20. Is there anything else about seasonal illness or local response you would like to share?

Appendix B. Key-Informant Interview Record

This appendix lists the interview domains used during the field visit. The full question-and-answer record is presented in Chapter 8 to keep the main narrative close to the field context.

Domain	Questions covered
Seasonal chronology	Seasonal shifts; differences between dengue and malaria timing; gastrointestinal illness timing.
Diagnosis and reporting	Rapid tests and laboratory confirmation; facility registers; reporting to the district surveillance unit.
Outbreak response	Local cluster recognition; coordination with sanitation teams; chlorine/ORS/vector-control response.
Operational constraints	Rapid-test supply; confirmatory laboratory turnaround; care-seeking before reaching public facilities.
Foodborne illness	Shared-meal clusters versus cases distributed over time; food safety during hot months.
Data-system improvement	Digital entry at triage, automated thresholds and integration with laboratory results.

Appendix C. Observation Checklist and Summary

I structured the site walk-through around four practical areas: outpatient flow, visible testing/support areas, facility access and the immediate surroundings. The summary below records the observations documented for the visit.

Area	Indicator	Summary recorded
OPD corridor	Patient flow and triage	High morning activity; manual workflow visible.
Support area	ORS and rapid-test materials	ORS corner and rapid-test materials described in field notes.
Facility boundary	Drainage and standing water	Intermittent stagnant water noted in nearby drainage.

Area	Indicator	Summary recorded
Residential surroundings	Water storage containers	Large plastic drums visible/described as household storage in the surrounding setting.
Communication	Public-health information	Public-facing setting was documented photographically; a systematic IEC poster count was not part of the recorded field note.

Appendix D. Field-Site Photographic Documentation

The photographs below document the site context supplied for this project. People in the busy corridor image have been visually softened to reduce identifiability. Captions describe only the visible setting.



Figure 10. Field-site photo panel: municipal exterior, covered entrance, outpatient corridor (faces softened) and surrounding roadside context.

Panel	What is shown
D1	Public-health facility exterior and signage.
D2	Covered public-facing entrance and approach.
D3	Outpatient waiting corridor; faces have been softened for privacy.
D4	Surrounding road, vegetation and drainage context visible in the supplied location image.

Appendix E. Summary Data Tables

The table below consolidates the principal numerical values used in the findings. Percentages refer to the event records or reported case values in the source analysis, not population incidence.

Family	Records	Cases	Deaths	Seasonal feature
Dengue	204	3,144	93	82.8% of events June–October; October peak
ADD	153	9,050	93	43.1% of events June–August; October secondary elevation
Malaria	70	2,133	51	51.4% of events May–July; 61.4% in Gadchiroli and Chandrapur
Food poisoning	69	5,927	29	Bimodal event distribution; case totals sensitive to three records
Chikungunya	43	701	0	Diffuse profile; Kendall W=0.150, p=0.832

Appendix F. Viva Preparation: Concise Responses

Question	Response
What is the project about?	I studied seasonal patterns in government-reported disease-outbreak records in Maharashtra using data processing and statistical analysis.
Why use NCDC/IDSP records?	They provide weekly reports with geography, disease labels and outbreak counts that can be compared over time.
What is the most important unit distinction?	One outbreak record is not one patient and does not measure population incidence.
Which disease showed the strongest late-year concentration?	Dengue: 82.8% of baseline events were reported from June through October, with October the largest month.
Why did I analyse food poisoning outliers?	Three high-volume records contributed 1,615 cases and materially changed February case totals.
Why is 2026 not compared as a full year?	The analysis stops at Week 32, before much of the late-monsoon/post-monsoon period.
What did the field visit add?	It provided context about facility workflow, diagnostic support, environmental setting and the difference between daily clinical activity and weekly outbreak notification.
What is one major limitation?	The source captures notified outbreak clusters, lacks population denominators and cannot establish causal effects from seasonality alone.
Why were the five disease families selected?	The repository audit selected families with sufficient baseline event counts and relevance to seasonal comparison.
Why keep raw disease labels?	Keeping source strings preserves traceability and lets later reviewers check whether normalization decisions were reasonable.
Why use an outlier sensitivity check?	It demonstrates whether the headline monthly case peak depends heavily on a few very large records.
What are my main recommendations?	Use disease-specific timing rather than one single “disease season,” show event counts separately from case volumes, extend the baseline as reports accumulate, and collect documented field evidence across more than one locality.